

Financial Econometrics

Course Recap

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This lecture

- ▶ Course recap
 - ▶ Learning goal
 - ▶ Lectures
- ▶ Coding cheat sheet
- ▶ Course evaluation
- ▶ Information on exam
- ▶ Q&A

Course in a nutshell

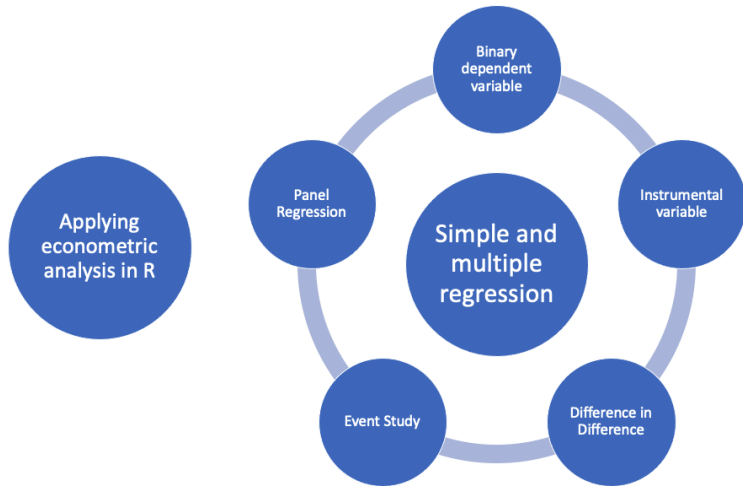
$$y = \alpha + \beta x + u$$

- ▶ Using data to measure a causal effect in a Financial context

Main teaching goal

- ▶ Given a question, you are able to...
 - ▶ choose the right model
 - ▶ gather the right data
 - ▶ implement the model
 - ▶ assess the quality of your analysis
- ▶ Econometric analysis is messy and there are many choices involved. You have to be certain that you can make decisions based on your analysis.
- ▶ Exam follows this principle.

Course structure



Course structure

- ▶ Introductory course - many different statistical tools
- ▶ For each subjects you have at least one (mostly two) fully coded examples readily available

R as a programming language

- ▶ R is an open source and widely used programming language
- ▶ “Most” popular language for data analysts
- ▶ Not only used for statistics but very flexible language
- ▶ Programming skills are transferable
 - ▶ Python (open source) and Matlab (commercial)
- ▶ There is no unique coding solution

Coding skill set

- ▶ Able to read code in class and adjust for your purpose
- ▶ Relevant tasks
 - ▶ loading data
 - ▶ merging data
 - ▶ constructing variables
 - ▶ select subset of data
 - ▶ estimate models
 - ▶ predict given a model
 - ▶ hypothesis testing

Steps in empirical economic analysis

1. Careful formulation of research question
2. Construct economic model
3. Turn into econometric model
 - ▶ specify econometric model
 - ▶ variable choices
 - ▶ hypothesis development
 - ▶ data gathering
 - ▶ estimation of model parameters

Simple regression model

$$y = \alpha + \beta x + u \quad (1)$$

- ▶ $\beta = \frac{\Delta y}{\Delta x}$ = slope coefficient; if x increases by one unit y increases by β units
- ▶ Ordinary Least Squares (OLS) minimizes the sum of squared residuals:

$$\hat{u}_i = y_i - \hat{y}_i = y_i - \hat{\alpha} - \hat{\beta}x_i$$

$$\min \sum_{i=1}^n \hat{u}_i^2 \rightarrow \hat{\alpha}, \hat{\beta}$$

$$\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$$

Why multiple regression?

$$y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$$

- ▶ Incorporate more explanatory factors into the model to avoid omitted variable bias
- ▶ Explicitly hold fixed other factors
- ▶ Allow a more flexible functional form

Variable selection

- ▶ Good control variables
 - ▶ If included in the regression, makes the error term uncorrelated with the variable of interest
 - ▶ Holding constant the control variable(s), the variable of interest is as if randomly assigned
- ▶ Protocol
 - ▶ Identify the variables of interest
 - ▶ Think of the omitted causal effect that could result in omitted variable bias
 - ▶ Include those omitted causal effects if you can or, if you cannot, include variables correlated with them that serve as control variables
 - ▶ This results in a “base” or “benchmark” model
 - ▶ Estimate your base model and plausible alternative specifications (robustness checks)

What to do with outliers?

- ▶ Drop
 - ▶ It is not acceptable to drop an observation just because it is an outlier
 - ▶ It is important to investigate the nature of the outlier before deciding
- ▶ Winsorize
 - ▶ Winsorizing or winsorization is the transformation of statistics by limiting extreme values to reduce the effect of possible spurious outliers
 - ▶ A typical strategy is to set all outliers to a specific percentile of the data: a 90% winsorization would see all data below the 5th percentile to the 5th percentile and data above the 95th percentile set to the 95th percentile.
- ▶ Transform the data (use logarithm for example). But be careful: interpretation of the slope will change.

Hypothesis testing

- ▶ Hypothesis testing
 - ▶ on a single parameter
 - ▶ jointly for several parameter
 - ▶ single restriction of multiple coefficients

Hypothesis testing - must know

- ▶ Use the sample to learn about the population
- ▶ We are not in the business of proving things
- ▶ p -value: acceptable probability of a type one error (rejecting the null hypothesis even if it is true in the population)
- ▶ t -statistic = $\frac{\hat{\beta} - \beta_{H0}}{St.Err.}$

OLS assumptions

- ▶ LS.1 The conditional distribution of u given x has mean zero, that is $E(u_i|x_i) = 0$
- ▶ LS.2 (y_i, x_i) are i.i.d.
- ▶ LS.3 No outliers
- ▶ LS.4 No perfect multicollinearity

If all OLS assumptions are satisfied: properties of $\hat{\beta}$

- ▶ Unbiased: $E(\hat{\beta}_0) = \beta_0$ and $E(\hat{\beta}_1) = \beta_1$
- ▶ Consistent: $plim(\hat{\beta}_0) = \beta_0$ and $plim(\hat{\beta}_1) = \beta_1$
- ▶ Normally distributed: $\hat{\beta}_1 \sim^a N(\beta_1, Var(\hat{\beta}_1))$

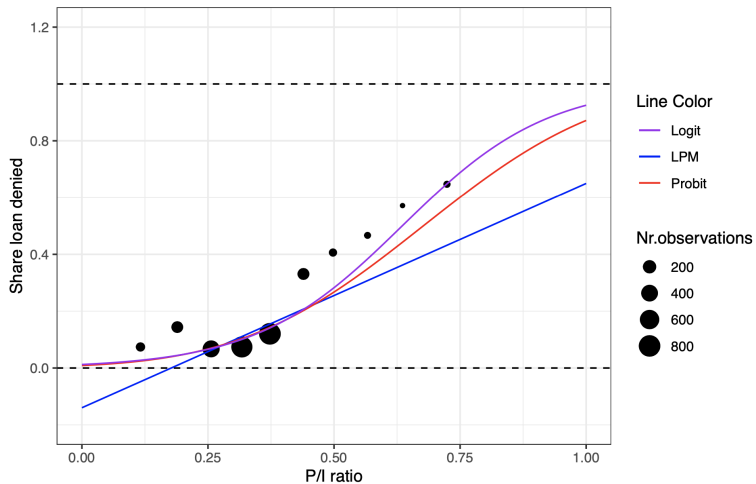
Usual suspects that lead to violation of LS.1

- ▶ Sample selection bias
- ▶ Omitted variable bias
- ▶ Simultaneity bias (incl. reverse causality bias)
- ▶ Measurement error in independent variable
- ▶ Small trick to find what might bias your analysis: imaging the perfect experiment. How does reality differ?

Binary dependent variables

- ▶ The dependent variable takes either 1 or 0
- ▶ Three options
 - ▶ Linear probability model
 - ▶ Probit
 - ▶ Logit

Binary dependent variables



Linear probability model (LPM)

- ▶ The LPM models $Prob(y = 1|x)$ as a linear function of x
- ▶ Advantages:
 - ▶ simple to estimate and interpret
 - ▶ inference the same as for multiple regression
- ▶ Disadvantages:
 - ▶ A LPM says that the change in the predicted probability for a given change in x is the same for all values of x , but that does not always make sense
 - ▶ Also, LPM predicts probabilities that can be < 0 and > 1

Logit and Probit regression

- ▶ Linking function
- ▶ Probit regression models the probability that $y = 1$ using the cumulative standard normal distribution, $\Phi(z)$ evaluated at $z = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$

$$Prob(y = 1|x) = \Phi(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$$

- ▶ Logit regression models the probability that $y = 1$ using the cumulative standard logistic distribution function, $\Lambda(z)$ evaluated at $z = \alpha + \beta_1 x_1 + \cdots + \beta_k x_k$

$$Prob(y = 1|x) = \Lambda(\alpha + \beta_1 x_1 + \cdots + \beta_k x_k)$$

Logit and Probit regression

- ▶ Because logit and probit use different probability functions, the coefficients (β 's) are different in logit and probit
- ▶ Coefficients hard to interpret $\Rightarrow$ use marginal effects
- ▶ If you choose to use logit / probit, rather report marginal effects than coefficients!

Panel data

- ▶ Follow entities over time
- ▶ With panel data we can control for unobserved factors that:
 - ▶ Vary across entities but do not vary over time
 - ▶ Vary across time but do not vary over entities
- ▶ Not accounting for them will lead to an omitted variable bias
- ▶ As aforementioned are unobserved, they cannot be included in a multiple regression

Event study

- ▶ Analysis of whether there was a statistically significant reaction in financial markets to past occurrences of a given type of event that is hypothesized to affect firm's market value
- ▶ Crucial elements:
 1. Identification of event
 2. Estimation and event windows
 3. Compute abnormal returns
 4. Compute cumulative abnormal return
 5. Hypothesis testing

Causality: Instrumental variable

- ▶ If **Zero Conditional Mean** assumption is violated, we have an endogeneity problem (coefficients are biased)
- ▶ Solution: break x in two parts: a part that is correlated with u , and a part that is not.
- ▶ Two Stage Least Squares (2SLS) estimation:
 - ▶ First Stage: regress endogenous regressor x on instrument z and controls w
 - ▶ Second Stage: use predicted $\hat{x}$ in original regression

Valid instrumental variable

- ▶ Relevance: Show that z explains x with an F -statistic > 10 (see multiple hypothesis testing) in the first stage regression
- ▶ Exogeneity: J -test for exogeneity, but mostly argumentative.

Causality: experiment and quasi-experiment

- ▶ An **experiment** is designed and implemented consciously by human researchers. An experiment entails conscious use of a treatment and control group with random assignment
- ▶ A **quasi-experiment** or **natural experiment** has a source of randomization that is “as if” randomly assigned, but the variation was not part of a conscious randomized treatment and control design

Difference in difference

$$y_{i,t} = \alpha + \beta_1 D_t \times G_i + \beta_2 D_t + \beta_3 G_i + u_{i,t}$$

- ▶ Treatment dummy D_t : turns one before treatment and zero afterwards
- ▶ Treatment group G_i : turns one if for treatment group and zero for control group
- ▶ Interaction term $D_t \times G_i$: effect of being in the treatment group after treatment was received
- ▶ β_1 is the difference in difference (DiD) estimator
- ▶ Construct counter-factual: what would have happened to the treatment group in the absence of treatment
- ▶ Parallel trend assumption requires that in the absence of treatment, the difference between the treatment and control group is constant over time

Matched control sample

- ▶ Nearest neighbor
 - ▶ For each treatment unit, we select a control firm that has the smallest matching error
 - ▶ Matching error is the distance between a pair of firms given selected characteristics
- ▶ Propensity score
 - ▶ We model the probability of being treated with a logit model
 - ▶ Subsequently, we select for each treated firm a control firm with a similar predicted probability of being treated

Coding cheat sheet

- ▶ Coding cheat sheet is available on CANVAS

Course Evaluation

- ▶ Please login to CANVAS and fill in course evaluation form

Final exam

- ▶ Written report (60% of the final grade)
 - ▶ hand-out 09:00 on 11.05.2026, submission 14:00 on 13.05.2026
- ▶ Oral exam (40% of the final grade)
 - ▶ held on 02.06.2026 and 04.06.2026 (in person, Aud B)

Previous exams

- ▶ You can find two examples of the previous exams on CANVAS in Files/Previous Exams folder.
- ▶ You can find a list of oral exam topics on CANVAS in Files/Oral exam folder.

Exam – written part

- ▶ Work in groups of 3-4 people
- ▶ You get a research question
- ▶ You get an intro and some implicit suggestions of how to approach this question
- ▶ You get a main data frame and several additional disjunct ones which you can use for additional variables
- ▶ You get very tight formal requirements
- ▶ You will have to make informed choices
- ▶ Results depend on choices and motivation behind the choices
- ▶ **Grades are group-based**
- ▶ **Grades can be repealed**

Formal requirements written part

- ▶ Your report should read like a mini thesis/article having following structure
 1. Abstract - summarize your findings (word limit = 100 words)
 2. Intro - outline the research objective
 3. Data - summarize the data used
 4. Analysis - your analysis
 5. Conclusion - your findings distilled
- ▶ The report will include three tables with column limits
- ▶ Word limit on writing (excl. abstract & tables)
- ▶ No appendix
- ▶ Make sure that tables are in line with guidelines on how to make proper tables
- ▶ Use fairly plain formatting ensuring readability
- ▶ Avoid redundancies or filler text
- ▶ You do not need to explain the empirical models used
- ▶ Efficiency of your coding will, given a lower bar, not be graded

Exam – oral part (i)

- ▶ Please sign up for a time slot on Canvas under People tab.
- ▶ 5 minutes presentation and approximately 20 minutes Q&A (30 minute slots)
- ▶ Please keep your answers short; I might interrupt so that we make most of the short time we have
- ▶ Centered on understanding of the Econometrics
- ▶ Covers the whole course. Focus on intuition.
- ▶ Given your answers, I will ask harder or easier follow up questions
- ▶ **Grades are individual**
- ▶ **Grades cannot be repealed**
- ▶ Objectivity ensured through TA presence and pre-set catalogue of relevant topics for discussion

Formal requirements: presentation

- ▶ The main purpose of the presentation is to be a starting point for the oral exam and it shall include your analysis
- ▶ The time limit of 5 minutes will be strictly enforced
- ▶ All three tables of the report should be in the presentation
- ▶ The formatting of the presentation, as long as the tables follow the aforementioned formatting rules, and your presentation skills, given a lower bar, are not graded
- ▶ You do not need to prepare material as potential answer to questions

Experiences from earlier years

- ▶ Try not to actively take away questions from your peer
- ▶ If you are the one knowing all answers, you might be the one getting the least questions (And that is a good thing)
- ▶ Really try to answer short. If I want to go into detail, I will
- ▶ Focus on intuition

Office hours

- ▶ Tuesday, 21.04.2026 from 14:00 till 16:00
- ▶ First come, first served

Any questions?

Good luck!